MI-19: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. Detailed review and scope.

1 MI-19: a rank-two counterexample to the subset inequality for the q-permanent

For $A = (a_{ij}) \in \mathbb{M}_n$ and $q \in \mathbb{R}$, use the inversion convention

$$\mathrm{per}_q(A) = \sum_{\sigma \in S_n} q^{\ell(\sigma)} \prod_{i=1}^{n} a_{i,\sigma(i)}, \qquad \ell(\sigma) = \#\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}.$$

For a subset $S \subseteq \{1, \dots, n\}$, define the restricted sum

$$R_{q,S}(A) = \sum_{\substack{\sigma \in S_n \\ \sigma(S) = S}} q^{\ell(\sigma)} \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

The target is $\mathrm{per}_q(A) \geq R_{q,S}(A)$ for Hermitian positive semidefinite A and $0 \leq q \leq 1$. This is the subset formulation recorded in [1, Conjecture 3, equation (5.1)].

Theorem 1.1. *The subset inequality is false for a real positive semidefinite matrix of order four and rank two. A counterexample is*

$$A = \begin{pmatrix} 170 & 1 & 167 & 170 \\ 1 & 1 & -2 & 1 \\ 167 & -2 & 173 & 167 \\ 170 & 1 & 167 & 170 \end{pmatrix}, \qquad q = \frac{7}{8}, \qquad S = \{2\}.$$

Proof. Positive semidefiniteness and rank two follow from

$$A = X^T X, \qquad X = \begin{pmatrix} 13 & 0 & 13 & 13 \\ 1 & 1 & -2 & 1 \end{pmatrix}.$$

Expansion over the 24 permutations gives

$$\begin{aligned} \mathrm{per}_q(A) = {} & 115600q^6 + 4886140q^5 + 9568969q^4 + 4712758q^3 \\ & - 199231q^2 + 4886140q + 4999700. \end{aligned}$$

The six permutations fixing the second index give

$$R_{q,\{2\}}(A) = 4999700q^5 + 9482260q^4 + 4741130q^3 + 4741130q + 4999700.$$

Consequently the difference is the small polynomial

$$\operatorname{per}_q(A) - R_{q,\{2\}}(A) = q(q+1)\big(115600q^4 - 229160q^3 + 315869q^2 \\ - 344241q + 145010\big).$$

At $q = 7/8$ this equals

$$-\frac{3235575}{16384} < 0.$$

All calculations are integer polynomial arithmetic. The accompanying verifier independently enumerates the permutations with the stated inversion convention. ☐

Remark 1.2 (Positive definite variants and scope). The strict negative gap persists for $A + \varepsilon I_4$ for every sufficiently small positive ε, by continuity of the two polynomial expressions. Thus the obstruction is not dependent on allowing a singular input. No counterexample to the separate monotonicity problem MI-18 is asserted.

References

[1] Carlos M. da Fonseca. The μ-permanent revisited. arXiv:1804.02231, 2018. Conjecture 3, equation (5.1). URL: https://arxiv.org/abs/1804.02231.